

XGBOOST

XGBoost → Extreme gradient boosting

⇒ XGB is similar to gradient boosting

Gradient Boosting:

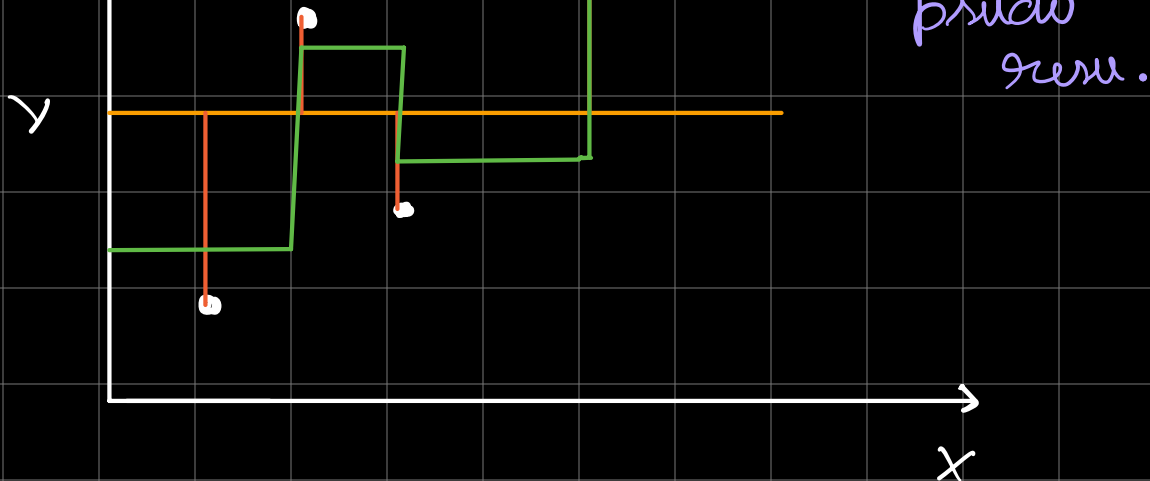
- 1) The first prediction is the mean of all the data.
- 2) The errors are called pseudo-residuals.



Stage 2) Train a decision tree

⇒ Input the data to predict the residual (error)

↑ } → new



Stage 3) train another decision tree to reduce the error even more.

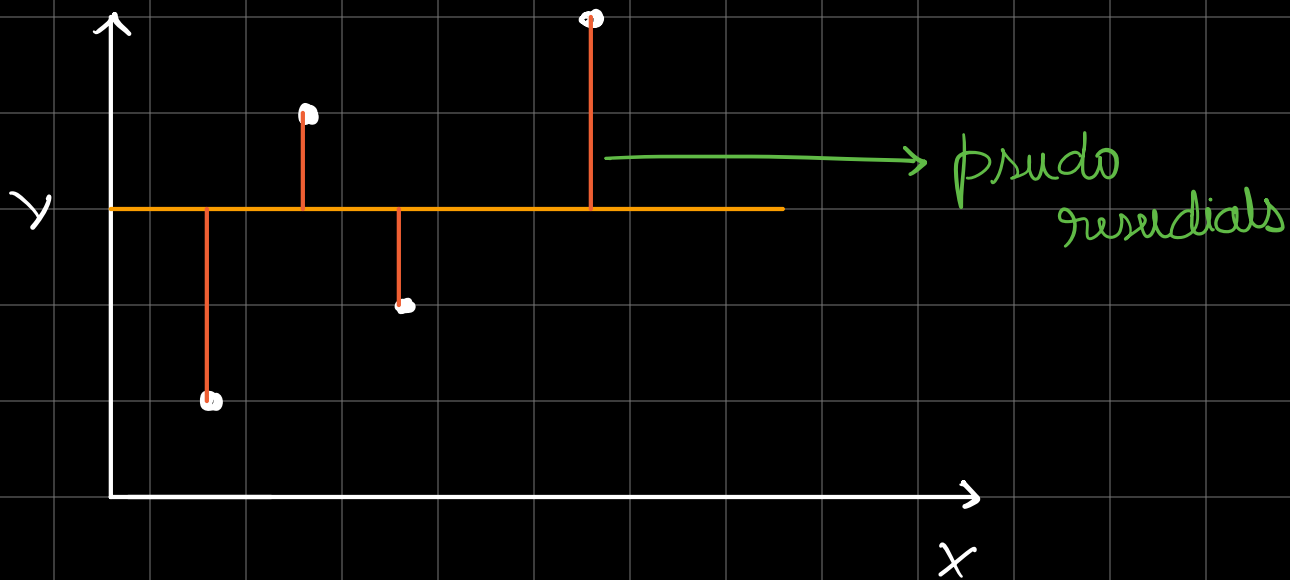
⇒ Decision trees are added stage wise to reduce errors / residuals.

GB
↓
vanilla
decision
trees
↓
Gini/entropy
is used

XGB
↓
In XGB the flow
of performing tasks is
similar to GB.
But the decision trees are not
vanilla dt's.
They don't use Gini/entropy.

XGBoosting:

Stage 1) Find the mean of the data.



x	y	R_i
6.7	4.5	-2.8
9.0	11.0	3.7
7.5	6.0	-1.3
5.0	8.0	0.7

residual 1
(Creates a new column of residuals)

used to
make DT

- ⇒ Make a leaf node to store all residuals.
- ⇒ Use this to find similarity score

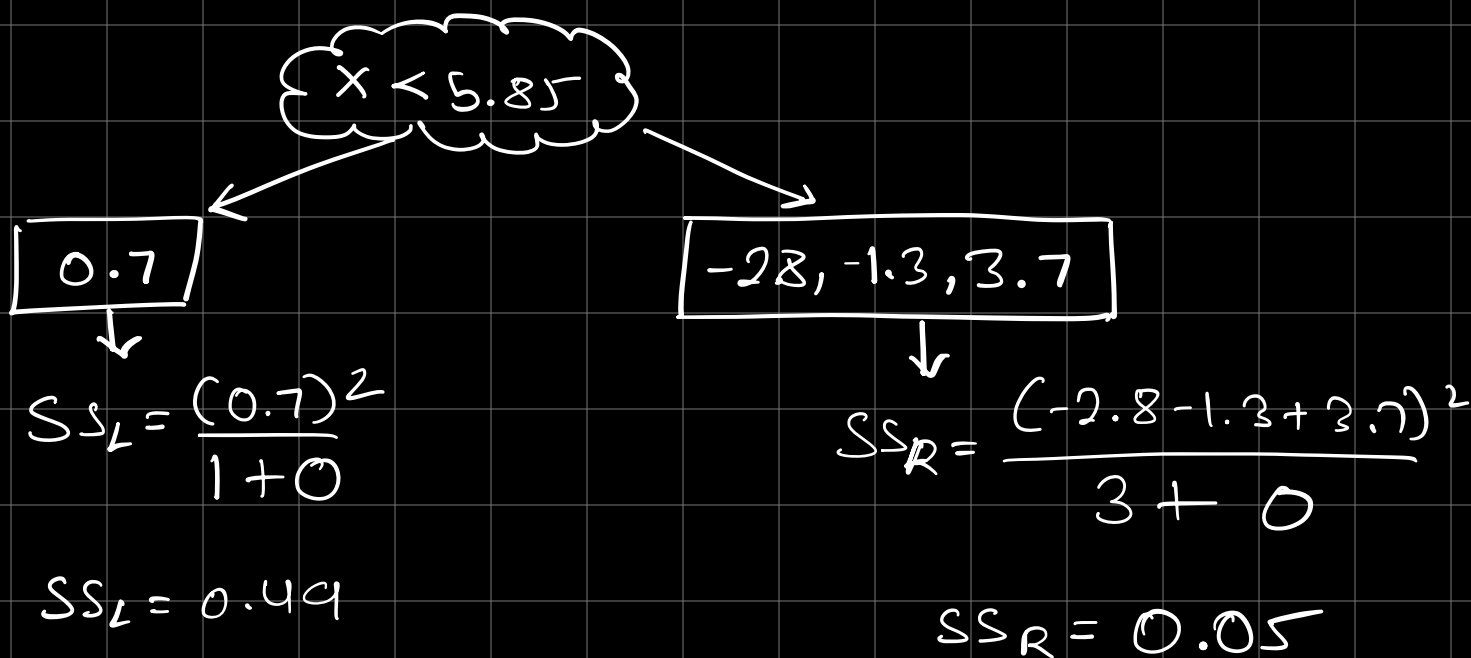
$$SS = \frac{(\text{sum of residuals})^2}{\text{no. of residual} + \lambda}$$

regularization
parameter

⇒ Now we try to split the residuals.
Splitting is done by using $\bar{x}$.



Splitting criteria ①, (5.85)



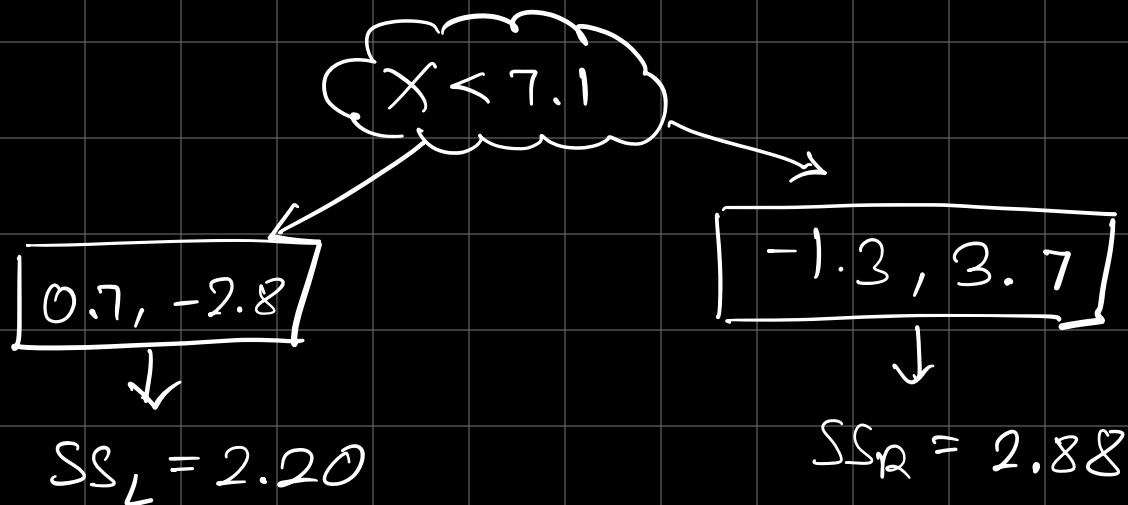
$$\{ \text{GAIN} = (SS_L + SS_R) - SS_{\text{ROOT}} \}$$

⇒ $SS \downarrow$ = residuals are similar

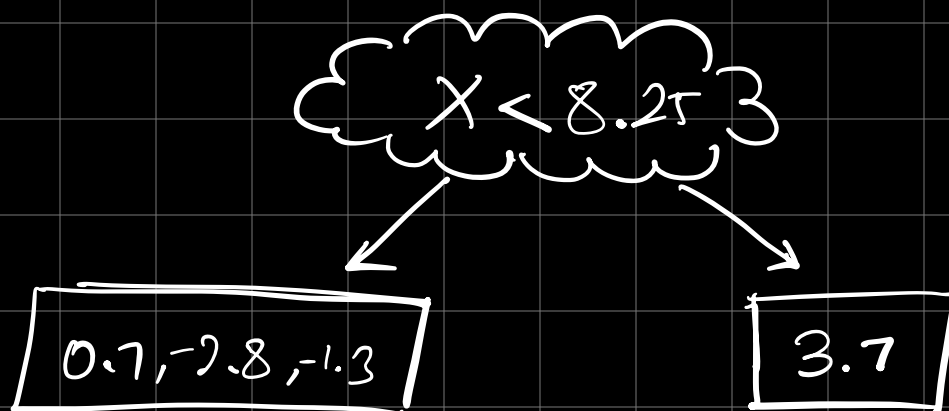
$\Rightarrow SS \uparrow$ = residuals are different

$\Rightarrow$ WE WANT TO MAXIMISE GAIN.

Splitting criteria (2), (7.1)

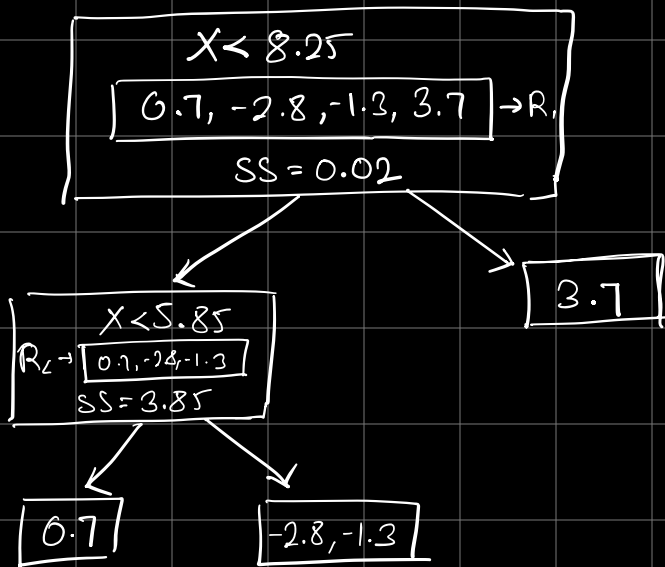


Stage 2] Make a decision tree with the splitting cri. as root node that make more. gain.

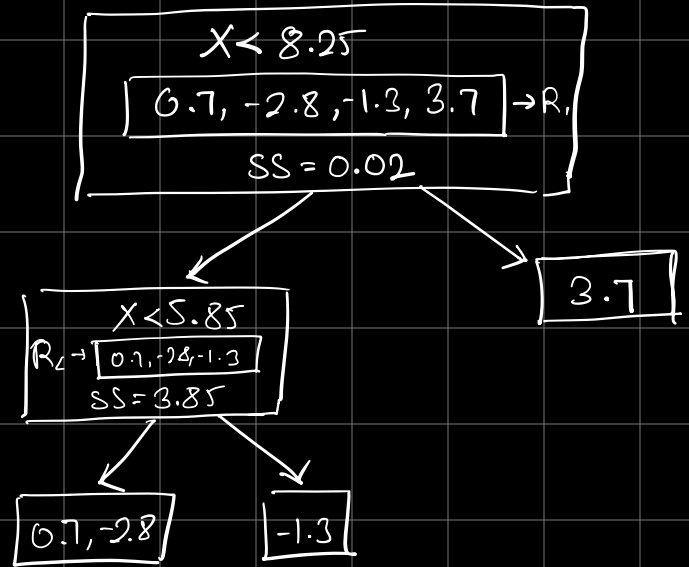


Stage 3) Split again based on the other splitting cri.

Splitting ①



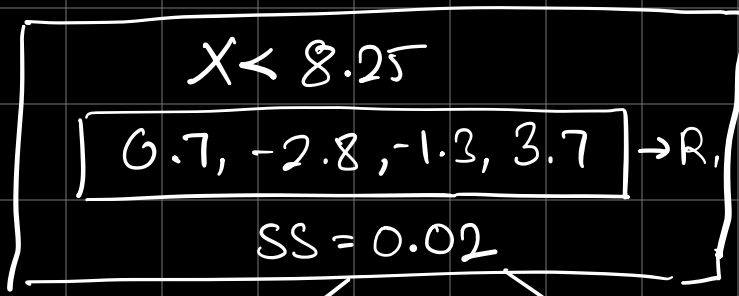
Splitting ②

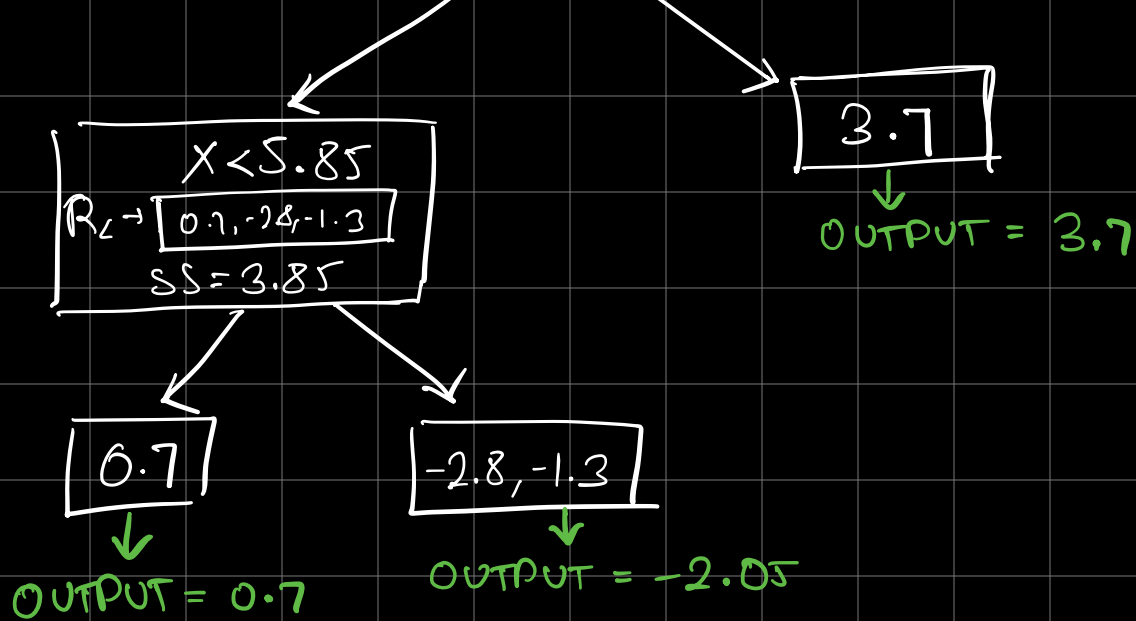


OUTPUT OF EACH LEAF NODE:

$$\text{OUTPUT} = \frac{\text{SUM OF RESIDUALS}}{\text{NO OF PARA.} + \lambda}$$

$\Rightarrow$ Now find value of output for each leaf.





x	y	R_1	P_1	P_2	R_2
6.7	4.5	-2.8	7.3	6.69	-2.19
9.0	11.0	3.7	7.3	8.41	2.51
7.5	6.0	-1.3	7.3	6.69	-0.69
5.0	8.0	0.7	7.3	7.51	0.49

$$\text{PREDICTED VALUE} = \text{BASE MODEL} + \sum L R \times M_i$$

⇒ GOAL IS TO REDUCE RESIDUALS TO 0

Another model will be trained with DT on X and R_2 . This cycle keeps repeating.

XGBOOST ON CLASSIFICATION

X	Y
5.70	0
6.25	1
7.10	0
8.15	1
9.60	1

Stage 1) Find the base model
 $\log(\text{odds}) = \log\left(\frac{P}{1-P}\right)$

P → Probability of +ve.

→ Eg: $\log(\text{odds}) = \log\left(\frac{3/5}{2/5}\right)$

X	Y	PRED1
5.70	0	0.405
6.25	1	0.405
7.10	0	0.405
8.15	1	0.405
9.60	1	0.405

$$\text{Probability} = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$

X	Y	PRED1	PROB
5.70	0	0.405	0.6
6.25	1	0.405	0.6
7.10	0	0.405	0.6

8.15	1	0.405	0.6
9.60	1	0.405	0.6

Errors of stage ①,

X	Y	PRED1	PROB	R _i
5.70	0	0.405	0.6	-0.6
6.25	1	0.405	0.6	0.4
7.10	0	0.405	0.6	-0.6
8.15	1	0.405	0.6	0.4
9.60	1	0.405	0.6	0.4

$$R_i = Y - \text{PROB}$$

LEAF NODE:

-0.6, 0.4, -0.6, 0.4, 0.4

Similarity
Score
in classification

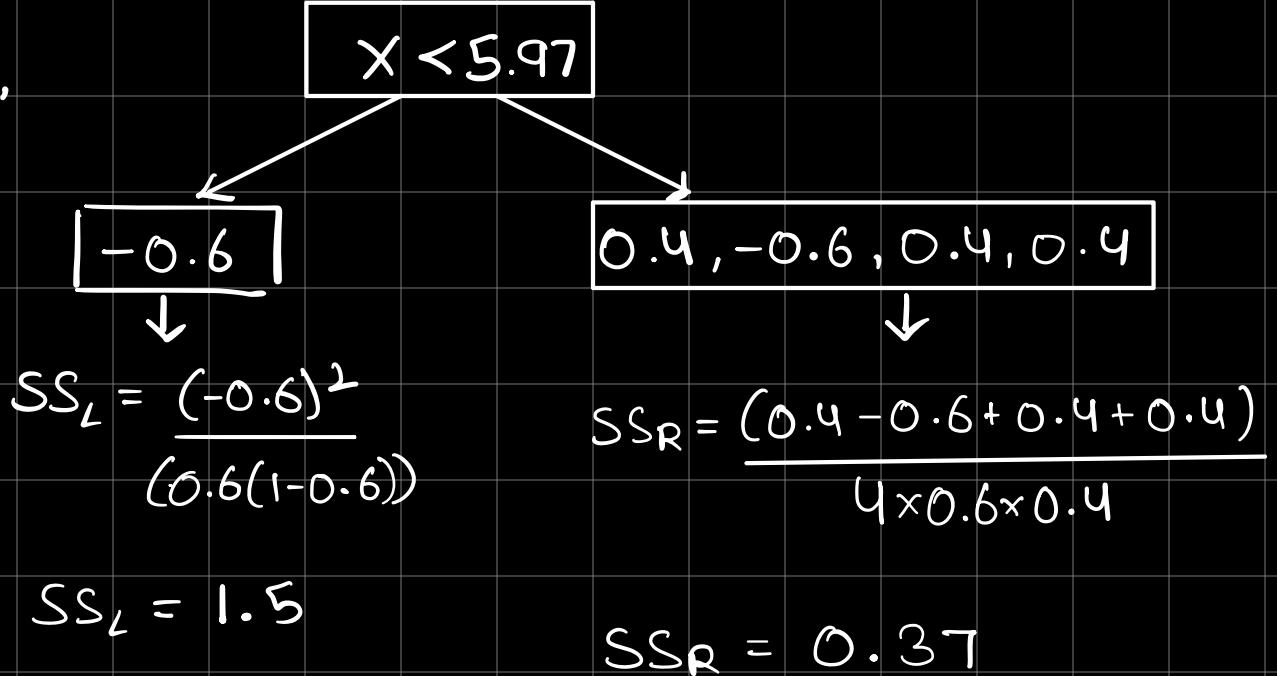
$$\Rightarrow SS = \frac{(\sum \text{Residual}_i)^2}{(\sum \text{Prev PROB}_i (1 - \text{Prev. PROB}_i) + \lambda)}$$

SPLITTING:

-0.6, 0.4, -0.6, 0.4, 0.4

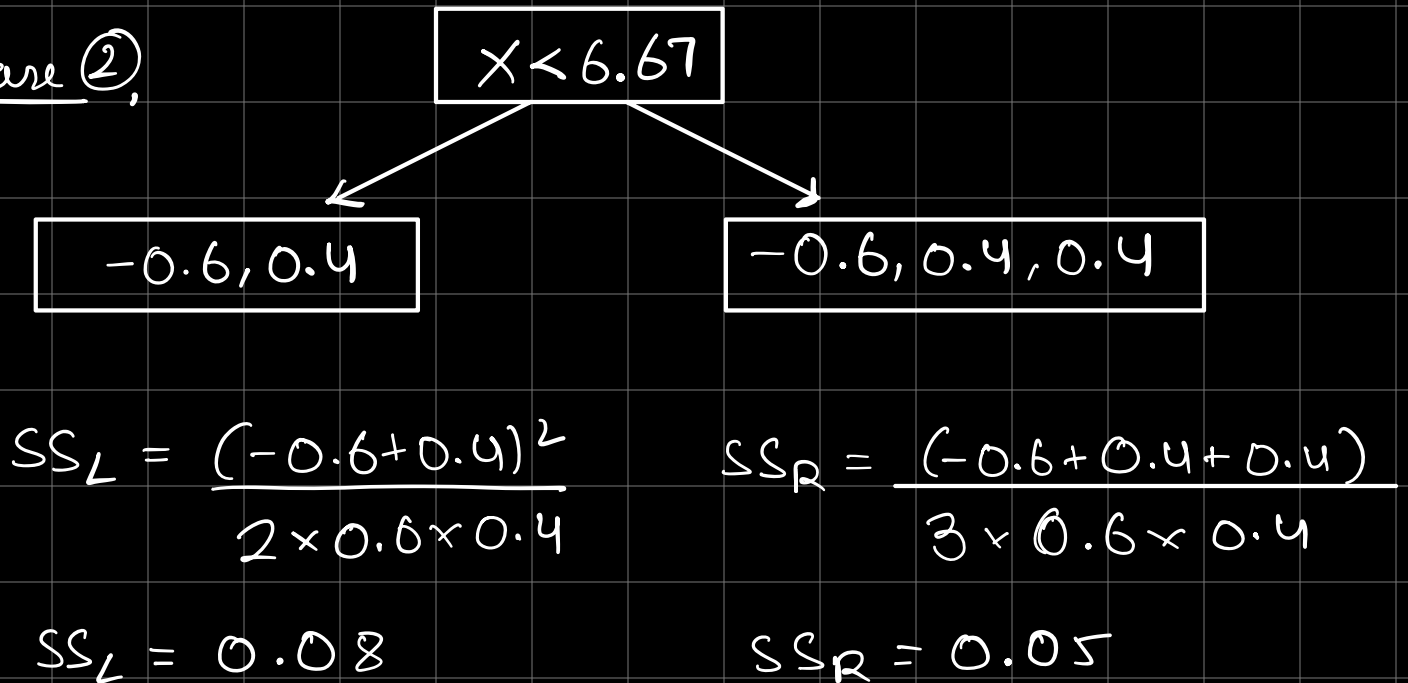
5.97 6.67 7.62 8.87

case ①,



GAIN $\Rightarrow SS_L + SS_R - SS_{ROOT}$
 $\Rightarrow (1.5 + 0.37) - 0$
 $\Rightarrow 1.87$

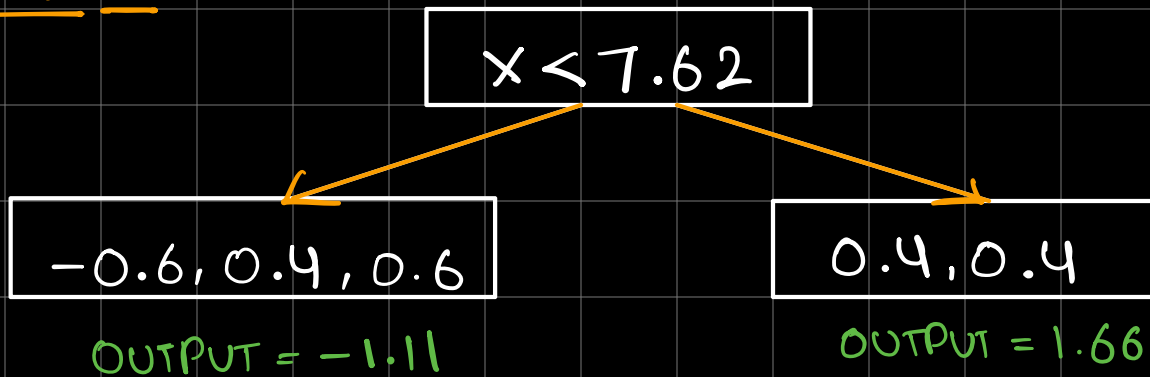
case ②,



GAIN $\Rightarrow SS_L + SS_R - SS_{ROOT}$
 $\Rightarrow 1.13$

Similarly find other nodes

FINAL DT:



$$\text{OUTPUT} = \frac{\sum \text{Residual}}{\sum \text{PrevP}(1 - \text{PrevP}) + \lambda}$$

$$\text{PREDICTED VALUE} = \text{BASE MODEL} + \sum \lambda R \times M_i$$